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**Department of Information Technology**

# Software Requirements Specification (SRS)

**Project Title:** Digital Library: Design and develop a portal to automate book cataloging, issue/return transactions, member management, online search, and fine calculation using network-based access, concurrent book transactions, and secure data management.

## Team Members:

| Roll No | Name of the student |
|---|---|
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# 1. Introduction

## 1.1 Problem Statement

Libraries in schools, colleges, and organizations often rely on outdated systems to manage digital assets and members. As educational paradigms shift toward remote and hybrid learning, these legacy systems are failing to meet modern accessibility demands. As the number of digital books and members increases, relying on raw file sharing without access tracking becomes inefficient and prone to security and copyright risks. Without secure file streaming protocols, institutions face rampant unauthorized distribution of copyrighted academic materials.

The problem exists because many libraries lack an integrated digital system that can automate catalog management, member records, online book searching, and secure digital transaction processing.

The people affected by this problem include:

- Librarians, who spend significant time maintaining digital access records and struggle to manually audit who is accessing specific files or investigate digital copyright breaches.
- Students and faculty members, who face delays in searching for literature and cannot synchronize their reading progress across devices. This fragmentation leads to a frustrating user experience, especially when switching between mobile, tablet, and desktop environments during study sessions.
- Library administrators, who find it difficult to monitor digital inventory and overall library usage, leaving them without the data-driven insights needed to justify budget allocations for future digital acquisitions.

A Digital Library Portal addresses these issues by providing a secure, network-based system that automates library operations while allowing multiple users to access the system simultaneously. By implementing "soft DRM" and role-based access, the portal bridges the gap between open academic access and strict digital security.

# 1.2 Need for the Project

A digital solution is required to improve the speed, accuracy, and reliability of library operations. Furthermore, with the rapid digitization of academic resources, institutions must actively protect their intellectual property while providing a seamless, 24/7 reading experience for their members.

### Current Challenges:

- Difficulty in searching books quickly across vast, unindexed digital archives.
- Duplicate or inconsistent member records due to decentralized or manual data entry processes.
- Limited access to library information outside of physical institutional networks or standard operating hours.
- Difficulty handling multiple users at the same time, often resulting in server crashes, race conditions, or access bottlenecks during peak exam seasons.
- Inability to securely control digital rights, leading to unauthorized file sharing and potential licensing violations.
- Complete lack of automated synchronization for user-specific data like bookmarks, page progress, and marginalia.
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### Motivation

The Digital Library Portal is designed to automate routine library activities using a centralized database and network-based access. It enables librarians and members to perform tasks efficiently while ensuring secure data management, concurrent digital transactions, and accurate record keeping.

By transitioning from a passive file repository to an active, secure streaming platform, the system prevents direct file downloads and protects valuable digital assets. The project also improves user experience through fast online searching and automated report generation, ultimately fostering a more engaging, highly available, and data-driven learning environment for the entire institution.

# 1.3 Objectives

The objectives of the Digital Library Portal are:

1. Automate book cataloging, issue, and return operations.
2. Provide fast online searching of books based on title, author, category, or ISBN.
3. Maintain secure and centralized member and book records.
4. Automatically calculate overdue fines and generate transaction reports.
5. Support concurrent access so multiple users can perform transactions simultaneously.
6. Improve the efficiency, accuracy, and reliability of overall library management.

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# 1.4 Scope

### In Scope (What the System Will Do)

- Maintain a centralized database of books and members.
- Add, update, delete, and search book records.
- Register and manage library members.
- Provide secure e-book file streaming (PDF/EPUB) to authenticated users.
- Display digital book availability and metadata in real time.
- Generate reports for accessed books, reading sessions, and user activity.
- Support multiple users through network-based access.
- Ensure secure login and data management using authentication.

### Out of Scope (What the System Will NOT Do)

- Online purchase or sale of books.
- Integration with external e-commerce websites.
- Mobile application development.
- AI-based book recommendations.
- Multi-library inter-library book exchange.

# 2. Problem Identification and Scope

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**Real Existing System Issues:**

- Legacy Integrated Library Systems (ILS): Systems like Koha are built for physical inventory. They treat digital files as external links, forcing users to leave the library portal to access content without native secure viewing.
- Open-Source Repositories (e.g., DSpace): Designed for archiving rather than active reading. Once a PDF/EPUB is downloaded, the system loses all tracking capability, leaving no audit trail for reading sessions.
- Commercial E-Book Platforms: Prohibitively expensive platforms that charge recurring per-user licensing fees and lock institutional data inside proprietary, closed-source ecosystems.

Limitations: No centralized database, no real-time search, no role-based access, manual tracking, no audit trail, no concurrent access support.

Problems Faced: Data errors, delayed transactions, lost records, poor digital tracking, no quick reporting, and digital copyright vulnerabilities.

**Proposed System**

- New Solution: Web-based, database-driven portal automating cataloging, digital transactions, progress tracking, and reporting with role-based access.
- Advantages: Centralized data, real-time search, automated reading analytics, concurrent transaction support, audit logs, and dynamic dashboards.
- Innovation: Thread-based concurrency handling, integrated analytics dashboard, and secure in-browser file streaming that prevents raw file downloads.

**Expected Benefits**

- Time Saving: Faster search, instant file access, and automated reporting.
- Cost Reduction: Less dependency on manual labor and expensive third-party commercial e-book licenses.
- Accuracy: 100% correct transactions and tracking, no duplicate/orphan data.
- User Convenience: Easy online search, real-time availability check, and self-tracking of cross-device reading history.

# 3. Feasibility Study

## 3.1 Technical Feasibility

The Digital Library Portal is technically feasible because it can be developed using widely available technologies and standard hardware. The system supports multiple users over a network, secure authentication, concurrent digital operations, and centralized database management.

| Component | Technology |
|---|---|
| Frontend | HTML5, CSS3, JavaScript, Bootstrap |
| Backend | JavaScript and Node.js |
| Database | MySQL |
| Web Server | AWS |
| Version Control | Git, GitHub |

| Tools | Purpose |
|---|---|
| Visual Studio Code | Source code development |
| AWS | Deploying Web Application and Database |
| Git and GitHub | Version control, Code repository and collaboration |
| Postman | Testing APIs |
| Web Browser | Overall web application testing |

## 3.2 Economic Feasibility

The Digital Library Portal is economically feasible because it uses open-source software and can be developed using existing computer systems. The implementation and maintenance costs are low, making it suitable for educational institutions.

## 3.3 Operational Feasibility

The Digital Library Portal is simple to operate and provides an intuitive web-based interface for librarians and members. Only minimal training is required:

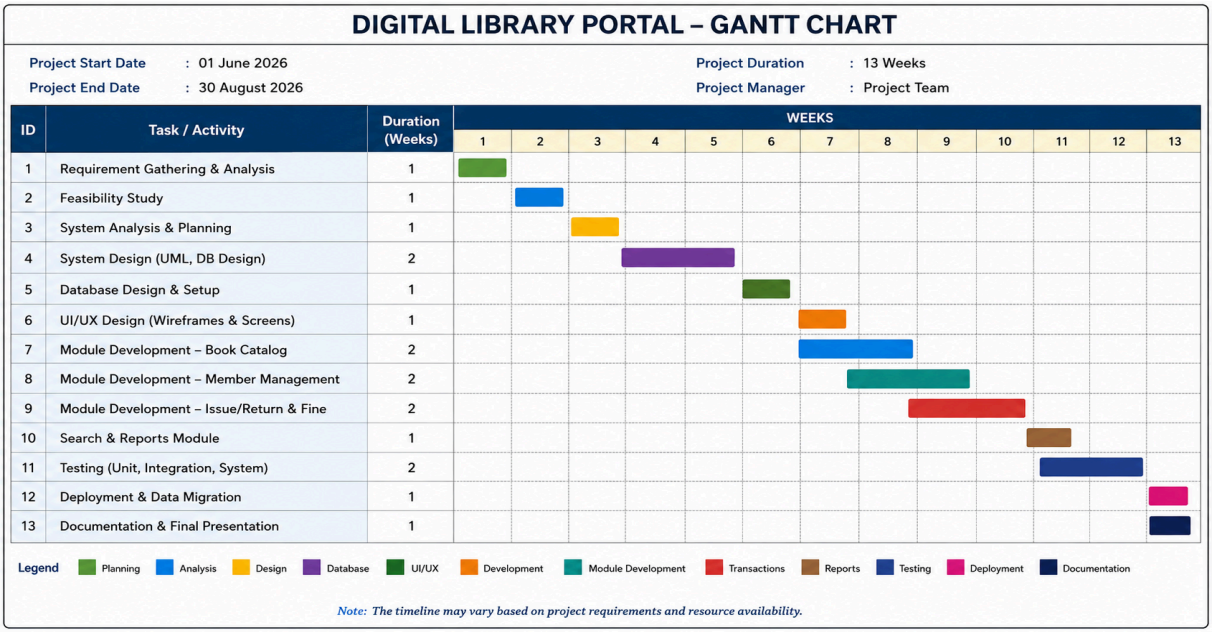
- Librarian: 1–2 days of training to manage books, members, and reports.
- Library Members: 30–60 minutes to learn searching, login, and account features.
- Administrator: Basic knowledge of database backup and user management.

## 3.4 Project Scope

The Digital Library Portal automates library operations by providing centralized management of books, members, transactions, and reports. It supports secure, network-based access and improves the efficiency and accuracy of library services.

**Deliverables**

- Software Requirements Specification (SRS)
- System Design (UML Diagrams)
- Database Design (ER Diagram and Tables)
- User Interface Screens
- Fully Functional Digital Library Portal
- Source Code



# 4. Stakeholder Analysis

| Stakeholder | Role | Expectations |
|---|---|---|
| Administrator (Admin) | Manage the overall library system, users, books, transactions, and system settings. | Easy management, secure access, accurate records, and automated reports. |
| Librarian | Catalog books, manage inventory, and oversee digital access logs. | Fast transactions, simple interface, accurate tracking, and reduced manual work. |
| Library Member (User) | Search, access, and read books; view reading history. | Fast and reliable access, real-time book availability, and user-friendly services. |
| Management | Monitor library operations, resource utilization, and overall performance. | Analytics, performance reports, inventory insights, and decision support. |
| System Administrator | Maintain the server, database, network, user accounts, and system security. | Reliable performance, secure data, regular backups, and minimal downtime. |
| Book Publisher | Provide new books and update catalog information when required. | Efficient inventory updates and smooth coordination with the library. |

# 5. Functional Requirements

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**FR1 (Registration):**
- The system shall allow administrators to register, update, and manage member profiles.

**FR2 (Authentication):**
- The system shall authenticate users securely and issue encrypted session tokens.

**FR3 (Authorization):**
- The system shall strictly restrict UI views and API endpoints based on role-based access control (RBAC).

**FR4 (Cataloging):**
- The system shall enable authorized staff to upload digital books and manage their metadata.

**FR5 (Categories):**
- The system shall support hierarchical categorization to organize the digital catalog efficiently.

**FR6 (Search):**
- The system shall allow members to instantly query the catalog by title, author, category, or ISBN.

**FR7 (Availability):**
- The system shall display real-time availability and descriptive metadata for all cataloged assets.

**FR8 (Streaming):**
- The system shall stream e-book files directly within the browser to prevent unauthorized raw file downloads.

**FR9 (Syncing):**
- The system shall automatically ping the server to save and synchronize reading progress during active sessions.

**FR10 (History):**
- The system shall maintain and display a persistent personal reading history for every member.

**FR11 (Dashboards):**

- The system shall generate role-specific graphical dashboards displaying real-time metrics.

**FR12 (Fines/Expiry):**

- The system shall automatically calculate and display overdue access flags and fines.

**FR13 (Reports):**

- The system shall allow administrators to dynamically generate and export library utilization reports.

**FR14 (Notifications):**

- The system shall push automated alerts for expiring digital leases and administrative broadcasts.

**FR15 (Auditing):**

- The system shall record an immutable audit log detailing every database modification.

**FR16 (Backups):**

- The system shall provide secure administrative tools to manually trigger database backups and restorations.

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# 6. Non-Functional Requirements

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**Performance**

- Response Time: The backend REST API shall process standard querying and CRUD operations with an average response time of <2 seconds.
- Throughput & Concurrency: The Node.js server shall utilize non-blocking I/O and connection pooling to support a minimum of 10 concurrent users actively streaming books without race conditions or memory leaks.
- Report Generation: Complex analytical queries and report generation shall complete and render in <5 seconds.

**Security**

- Authentication & Authorization: API endpoints shall be secured using HTTP Bearer Token authentication (JWT) with token expiration.
- Data Encryption: All user passwords shall be cryptographically salted and hashed (using algorithms like bcrypt) before database insertion; plaintext passwords shall never be stored.
- Vulnerability Mitigation: The application shall utilize parameterized SQL queries or an ORM to strictly prevent SQL Injection (SQLi) attacks. Cross-Site Scripting (XSS) prevention mechanisms shall be implemented on all frontend inputs.

**Reliability & Availability**

- Uptime Target: The application shall target a 99% uptime during standard institutional operating hours.
- Fault Tolerance: The system shall implement robust error handling (e.g., HTTP 4xx and 5xx status codes) to ensure that isolated module failures do not crash the primary server application.
- Data Preservation: Automated background jobs shall execute daily logical backups of the MySQL database.

**Scalability**

- Stateless Architecture: The Node.js backend shall be designed as a stateless application, allowing it to easily scale horizontally across multiple AWS instances as the user base grows.
- Storage Expansion: The database and file storage schemas shall be optimized to accommodate a continuous future increase in cataloged books and registered users without requiring schema refactoring.

**Maintainability**

- Code Quality: The source code shall adhere to a modular architecture separating routing, business logic (controllers), and database interactions (models).
- Standards: The development team shall follow standard JavaScript/Node.js naming conventions and utilize a RESTful API design philosophy for all network endpoints.

**Portability & Compatibility**

- Cross-Platform Client: The web portal shall execute correctly on modern, standard-compliant web browsers (Chrome, Firefox, Safari, Edge) independent of the user's underlying operating system.
- Server Environment: The backend runtime and database environment shall be containerizable or portable enough to deploy successfully on Windows, Linux, or cloud-based Web infrastructure (AWS).

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# 7. System Requirements

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## Hardware

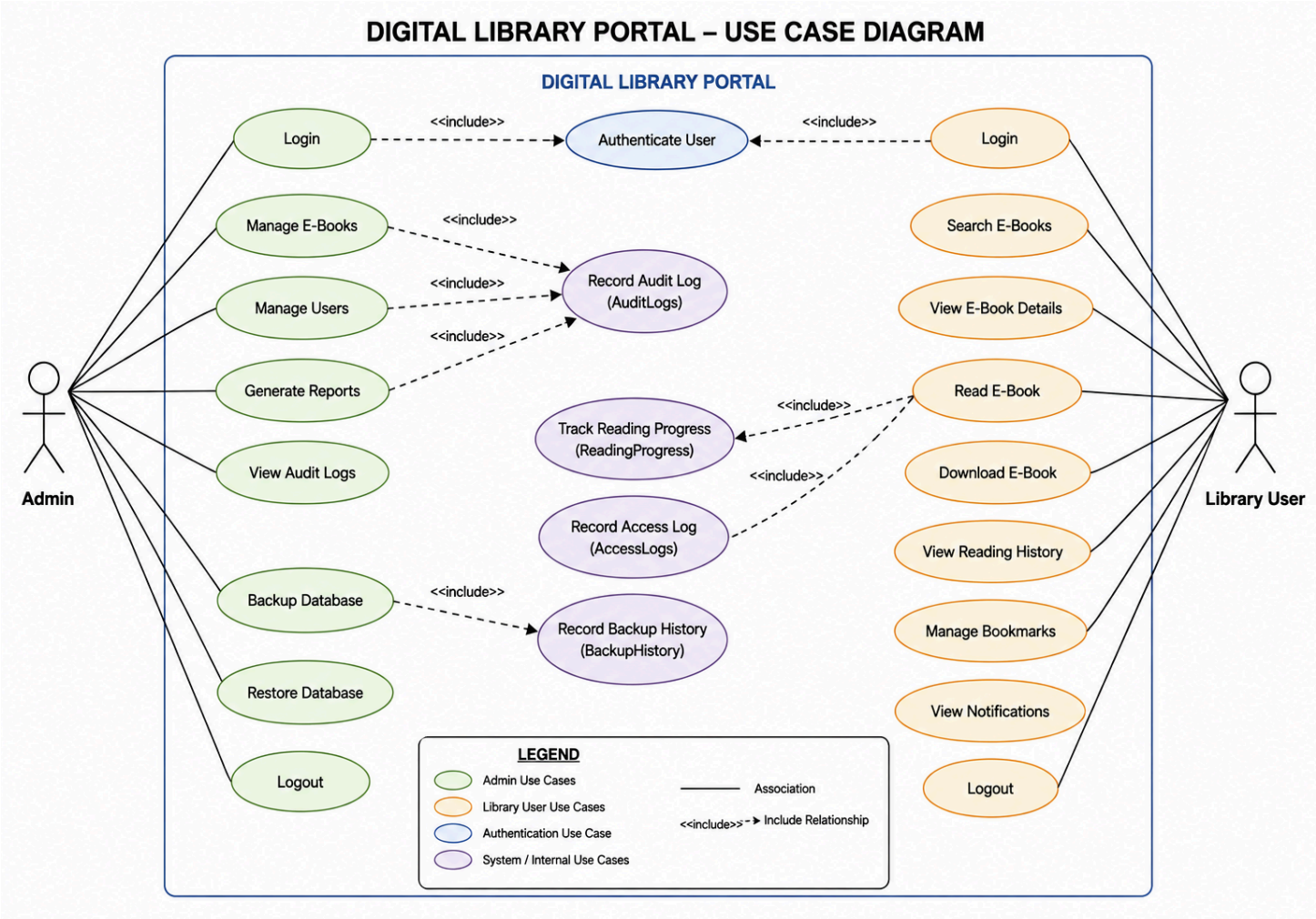
- 8 GB RAM
- Intel i5
- 256 GB SSD

## Software

- Windows/Linux
- VS Code
- MySQL
- NodeJS
- Browser

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# 8. Use Case Diagram



# 9. Use Case Descriptions

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### UC-01: Login

**Actor:** Admin / Library User
**Description:** Allows users to securely log into the Digital Library Portal.
**Precondition:** User has a registered account.
**Main Flow:** Enter credentials → System authenticates user → Access granted according to role.
**Alternative:** Invalid credentials → Display error message.
**Postcondition:** User is authenticated.

### UC-02: Manage E-Books

**Actor:** Admin
**Description:** Allows the administrator to add, update, delete, and manage e-book records and metadata.
**Precondition:** Admin is authenticated.
**Main Flow:** Select operation → Modify e-book information → System saves changes → Audit log recorded.
**Postcondition:** E-book records are updated.

### UC-03: Manage Users

**Actor:** Admin
**Description:** Allows the administrator to register and manage library user accounts.
**Precondition:** Admin is authenticated.
**Main Flow:** Select user → Add/update user information → System saves changes → Audit log recorded.
**Postcondition:** User records are updated.

### UC-04: Generate Reports

**Actor:** Admin
**Description:** Generates reports on library usage, reading sessions, accessed books, and user activity.
**Precondition:** Admin is authenticated.

**Main Flow:** Select report type → Retrieve data → Generate report.
**Postcondition:** Report is generated.

### UC-05: View Audit Logs

**Actor:** Admin
**Description:** Allows the administrator to view recorded system and database modification activities.
**Precondition:** Admin is authenticated.
**Main Flow:** Request audit logs → System retrieves records → Display logs.
**Postcondition:** Audit information is displayed.

### UC-06: Backup Database

**Actor:** Admin
**Description:** Allows the administrator to create a secure database backup.
**Precondition:** Admin is authenticated.
**Main Flow:** Select backup → System creates backup → Backup history recorded.
**Postcondition:** Database backup is stored.

### UC-07: Restore Database

**Actor:** Admin
**Description:** Allows the administrator to restore the database from an existing backup.
**Precondition:** Admin is authenticated and a valid backup exists.
**Main Flow:** Select backup → Validate backup → Restore database.
**Postcondition:** Database is restored.

### UC-08: Search E-Books

**Actor:** Library User
**Description:** Allows users to search e-books by title, author, category, or ISBN.
**Precondition:** User is authenticated.
**Main Flow:** Enter search query → System searches catalog → Display results.
**Postcondition:** Matching e-books are displayed.

### UC-09: View E-Book Details

**Actor:** Library User
**Description:** Displays detailed metadata and availability information for a selected e-book.
**Main Flow:** Select e-book → System retrieves information → Display details.
**Postcondition:** E-book details are displayed.

### UC-10: Read E-Book

**Actor:** Library User
**Description:** Allows authenticated users to read an e-book through the portal.
**Precondition:** User has access to the e-book.
**Main Flow:** Select e-book → Verify access → Stream e-book → Track progress and access.
**Postcondition:** Reading activity is recorded.

### UC-11: Stream E-Book

**Actor:** Library User
**Description:** Provides secure in-browser access to e-book content without exposing the raw file.
**Precondition:** User is authenticated and authorized.
**Main Flow:** Request e-book → Verify access → Stream content in browser.
**Postcondition:** E-book is securely displayed.

### UC-12: View Reading History

**Actor:** Library User
**Description:** Allows users to view their previous reading activity and progress.
**Main Flow:** Request history → System retrieves records → Display reading history.
**Postcondition:** Reading history is displayed.

### UC-13: Manage Bookmarks

**Actor:** Library User
**Description:** Allows users to create, view, update, and remove e-book bookmarks.
**Main Flow:** Select bookmark option → Add/update/delete bookmark → System saves changes.
**Postcondition:** Bookmarks are updated.

### UC-14: View Notifications

**Actor:** Library User
**Description:** Allows users to view system-generated notifications and alerts.

**Main Flow:** Open notifications → System retrieves notifications → Display messages.
**Postcondition:** Notifications are displayed.

### Included Use Cases

**UC-15: Authenticate User**
Validates user credentials and establishes an authenticated session.

**UC-16: Track Reading Progress**
Automatically records and synchronizes reading progress during an active session.

**UC-17: Record Access Log**
Records user access and reading activity.

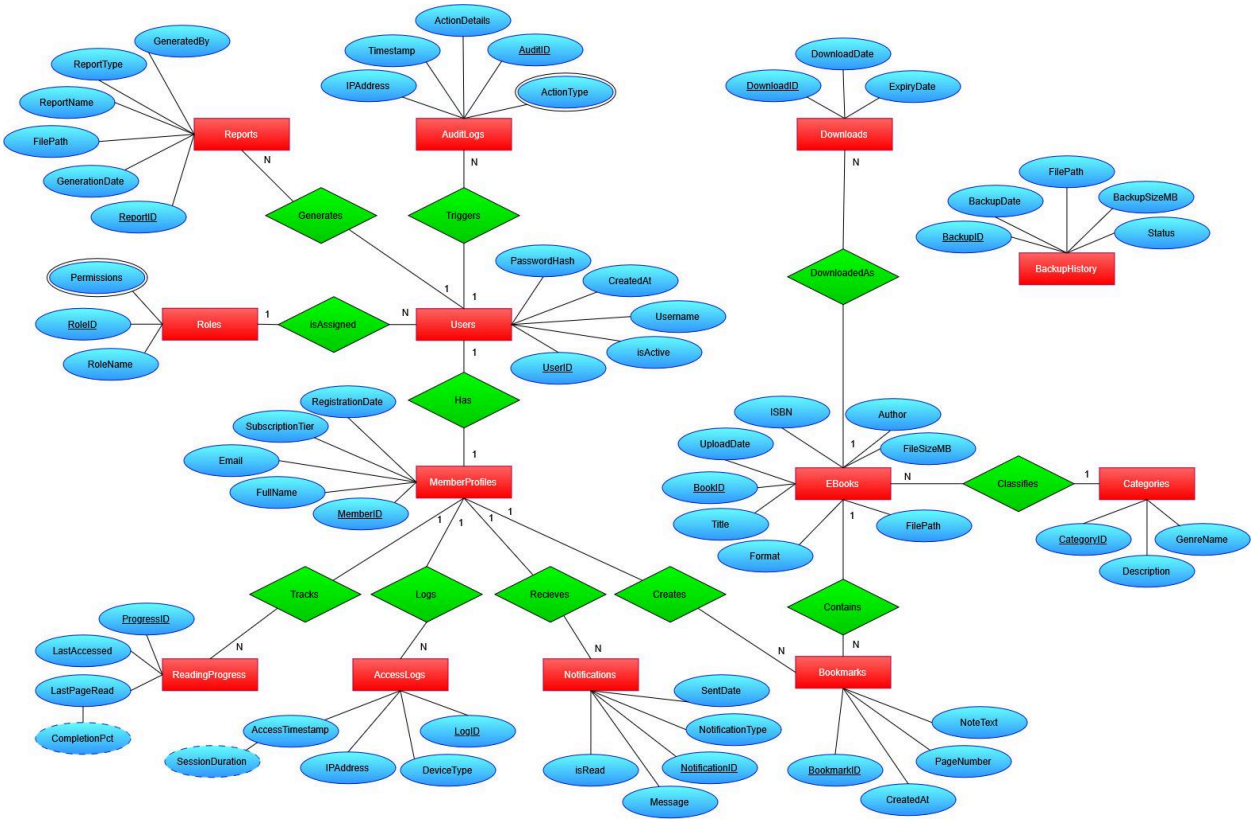
**UC-18: Record Audit Log**
Records database modifications and administrative actions.

**UC-19: Record Backup History**
Records completed database backup operations.

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# 10. Data Requirements

## 10.0 Entity Relationship Diagram



## 10.1 Database Tables

The Digital Library Port

al shall use a MySQL relational database containing the following tables:

1. **Roles** – Stores system roles and their permissions.
2. **Users** – Stores login credentials and user account information.
3. **MemberProfiles** – Stores additional information of library members.
4. **Categories** – Stores e-book categories and descriptions.
5. **EBooks** – Stores e-book metadata and secure file information.
6. **AccessLogs** – Records user access and reading sessions.
7. **ReadingProgress** – Stores the reading progress of members for each e-book.
8. **Bookmarks** – Stores bookmarks created by members.
9. **Downloads** – Stores authorized e-book download records and expiry information.
10. **AuditLogs** – Records important administrative and system activities.
11. **Reports** – Stores generated report information.
12. **Notifications** – Stores notifications sent to library members.
13. **BackupHistory** – Stores information about database backup operations.

## 10.2 Important Attributes

The main attributes of the database tables are:

- **Roles:**

  RoleID (PK), RoleName, Permissions

- **Users:**

  UserID (PK), RoleID (FK), Username, PasswordHash, IsActive, CreatedAt

- **MemberProfiles:**

  MemberID (PK), UserID (FK), FullName, Email, SubscriptionTier, RegistrationDate

- **Categories:**

  CategoryID (PK), GenreName, Description

- **EBooks:**

BookID (PK), CategoryID (FK), Title, Author, ISBN, Format, FilePath, FileSizeMB, UploadDate

- **AccessLogs:**

  LogID (PK), MemberID (FK), BookID (FK), AccessTimestamp, IPAddress, DeviceType, SessionDuration

- **ReadingProgress:**

  ProgressID (PK), MemberID (FK), BookID (FK), LastPageRead, CompletionPct, LastAccessed

- **Bookmarks:**

  BookmarkID (PK), MemberID (FK), BookID (FK), PageNumber, NoteText, CreatedAt

- **Downloads:**

  DownloadID (PK), MemberID (FK), BookID (FK), DownloadDate, ExpiryDate

- **AuditLogs:**

  AuditID (PK), UserID (FK), ActionType, ActionDetails, Timestamp, IPAddress

- **Reports:**

  ReportID (PK), GeneratedBy (FK), ReportName, ReportType, GenerationDate, FilePath

- **Notifications:**

  NotificationID (PK), MemberID (FK), Message, NotificationType, SentDate, IsRead

- **BackupHistory:**

  BackupID (PK), BackupDate, Status, FilePath, BackupSizeMB

# 11. Assumptions and Constraints

## 11.1 Assumptions

- Every user has a unique User ID and valid login credentials.
- Every e-book has a unique Book ID.
- Only authenticated users can access the portal.
- Users have a stable internet connection.
- The MySQL database is available during system operation.
- Administrators have appropriate permissions to manage system data.
- Users provide accurate information during registration and system usage.

## 11.2 Constraints

- The system requires network/internet connectivity.
- The system uses MySQL for database management.
- The system is accessed through a web browser.
- Access to system functions is restricted according to user roles.
- E-book files must be stored in the configured secure storage location.
- System performance depends on available server, network, and database resources.

# 12. Future Enhancements

- **Mobile Application:** Develop Android/iOS applications for easier access to the digital library.
- **Advanced Search:** Add filters, recommendations, and more advanced search capabilities.
- **AI-Based Recommendations:** Recommend e-books based on users' reading history and interests.
- **Enhanced Accessibility:** Add features such as screen-reader support, text-to-speech, and improved accessibility controls.

# 13. References

## 13.1 Websites and Technical Documentation


- **MDN Web Docs – HTML, CSS and JavaScript Documentation**
  Used as a reference for web-page structure, styling, scripting, and Web APIs.
  MDN Web Docs
- **MySQL 8.4 Reference Manual**
  Used as a reference for relational database design, SQL, constraints, and database management.
  MySQL 8.4 Reference Manual
- **Node.js Documentation**
  Used as a reference for server-side JavaScript and backend development.
  Node.js Documentation
- **Bootstrap Documentation**
  Used as a reference for responsive web design and frontend components.
  Bootstrap Documentation
- **W3C – Web Content Accessibility Guidelines (WCAG) 2.2**
  Used as a reference for accessibility considerations in the web interface.
  W3C WCAG 2.2


## 13.2 Research Papers


- Kroll, H., Kreutz, C. K., Jehn, M., & Risse, T. (2024). **"Requirements for a Digital Library System: A Case Study in Digital Humanities."**
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- Kreutz, C. K., Blum, M., Schaer, P., Schenkel, R., & Weyers, B. (2023). **"Evaluating Digital Library Search Systems by using Formal Process Modelling."**
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  Research Paper – arXiv
- Mukherjee, S., & Patra, S. K. (2023). **"Digital Library Initiatives in India: A Comprehensive Study."**
  Used as a reference for digital-library initiatives and challenges in the Indian context.
  Research Paper – arXiv

## 13.3 Books


- Pressman, R. S., & Maxim, B. R. **Software Engineering: A Practitioner's Approach.** McGraw-Hill Education.
- Silberschatz, A., Korth, H. F., & Sudarshan, S. **Database System Concepts.** McGraw-Hill.


## 13.4 APIs / Standards

- **MDN Web APIs Documentation** – Reference for browser APIs used in web application development.
  MDN Web APIs